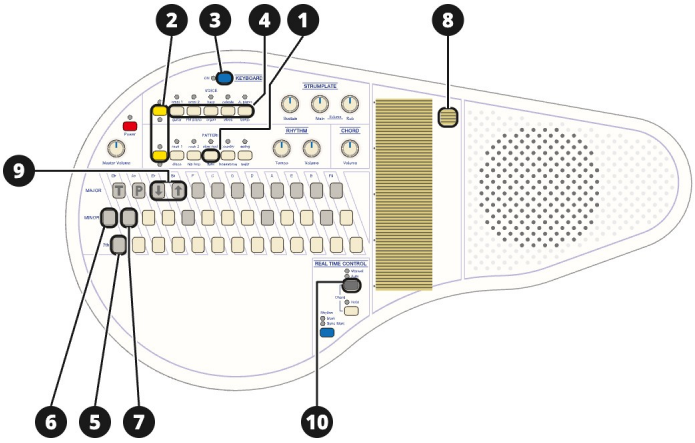
OM-108 CUSTOM FIRMWARE

1. **[PATTERN (slow rock/funk)] button** — Hold on startup to enter OM-27 rhythm mode. The 6 [PATTERN] buttons (including the yellow one) now play the 6 OM-27 rhythm patterns, left to right: Latin (yellow), Rock, Foxtrot, Slow Rock, Waltz, Swing. Hold any 2 [PATTERN] buttons and press the [KEYBOARD (ON)] button (3) to lock the selection.
2. **[VOICE] / [PATTERN] left buttons (yellow)** — Hold the [PATTERN] left button (yellow) on startup for an alternate (plucked) bass sound; hold the [VOICE] left button (yellow) for the stock bass sound.
3. **[KEYBOARD (ON)] button** — Hold on startup to enter Jazz Mode. See the table below for changed and new chord fingerings. This setting is not remembered between power cycles. The [KEYBOARD (ON)] button is also used as a function key for OM-27 mode (1) and the [Chord Manual/Auto] button (10).
4. **[VOICE] buttons** — After a voice has been selected, press the lit [VOICE] button again to toggle its alternate sub sound. "omni 1" has no alternate sub sound due to technical limitations.

Voice	omni 1	omni 2	harp	celeste	A. piano
Sub	–	Sine Pad	Dial Up	Sine Pad	Sine Pad
Alt	–	Mellow Pad	Evolve	Church Bell	Strings 2
Voice	**guitar**	**FM piano**	**organ**	**vibes**	**banjo**
Sub	Sine Pad	Mellow Pad	Dah Guys	Strings	Strings
Alt	Mellow Pad	Rezy	7th Doo	Hand Pan	Mellow Pad

5. **"G ♭ -7th" chord button** — In keyboard mode, this button sets the modulation mode of the Instant OFF plate.
   Tap: cycle to the next mode displayed on the [VOICE] LEDs. The device will announce the selected mode.
   Hold + [↑]/[↓] buttons (9): cycle through the modes silently.
   Hold + [VOICE] button 1–5 (4): choose one of the 5 modulation modes directly & silently.
6. **"G ♭ -MINOR" chord button** — In keyboard mode, hold to adjust the speed of the Auto modes with the [↑]/[↓] buttons (9), −6 to +6 on the [VOICE] LEDs.



7. **"D ♭ -MINOR" chord button** — In keyboard mode, hold to adjust the depth of the current mode, 0 to 12 on the [PATTERN] LEDs.
8. **Instant OFF plate** — using the settings described in (5), (6) and (7):
   - **Instant Off** (stock behavior)
   - **Vibrato Auto**: Like the stock behavior, a finger on the plate enables the vibrato set by (6) and (7), but the plate now detects partial coverage and adjusts speed and depth accordingly. Try rolling your finger off slowly after pressing to hear the effect.
   - **Vibrato Manual**: Rest a finger on a corner of the Instant OFF plate and rock it in and out, like bending a guitar string.
   - **Tremolo Auto**: The same as Vibrato Auto but for volume.
   - **Tremolo Manual**: The same as Vibrato Manual but for volume.
9. **[↓] / [↑] buttons ("E ♭ -MAJOR" / "B ♭ -MAJOR")** — These serve the same purpose for the modulation modes as they do for adjusting Transpose and Master Tune. Reminder: pressing both together resets the parameter.
10. **[Chord Manual/Auto] button** — Hold and press the [KEYBOARD (ON)] button at any time to enter a third Manual/Auto state combining Manual chords with the Auto bass (both LEDs lit).

Other Changes

- **Saved Settings:** All your settings are now saved between power-offs except for the new Jazz Mode.
- A factory reset (both [VOICE] and [PATTERN] left buttons (yellow) held while switching on) now sets the battery type to Ni-MH, and also clears the saved settings.
- The stock MIDI CC out messages (Volume & Release) are removed, making room for a fifth MIDI preset, P5.
- Two small bugs in the stock firmware, involving LED states and chord data sent via MIDI out, have been fixed.
- **Sounds:** The "harp" voice's main sound has been replaced with an electric harp sound, and the clap rhythm sound has been replaced.
- **Spoken Mode Names:** The unit announces the rhythm mode chosen at power-on ([PATTERN (swing/waltz)]: "84 mode", [PATTERN (country/bossanova)]: "108 mode", [PATTERN (slow rock/funk)] (1): "27 mode") and also the Modulation Mode selected via tapping.
- **Modulation Notes:** Both Vibrato modes send matching MIDI pitch bend out. Both Tremolo modes are local to the unit and are not sent over MIDI. Due to limitations of the analog circuits, neither Tremolo mode works on "omni 1" or the main voice of "omni 2".
- **Keyboard Mode Modulation:** In keyboard mode, hold a chromatic key and press the "G ♭ -MINOR" or "D ♭ -MINOR" chord button to play an Auto modulation mode on the played notes.

MIDI Presets Reference Table (choose at power-on: hold [VOICE] button 1–5). Sent on each part's channel:

Part	Channel	P1	P2	P3	P4	P5
Main (strumplate)	8			●	●	●
Chord	9	●		●	●	●
Bass	7				●	●
Sub (copy of Main)	9		●			
Drums *	1					●
Clock, Start / Stop	–				●	●

* Drum notes are sent on a smaller note range matching the Teenage Engineering OP-XY.

Jazz Mode Reference Table (shown in C); every root works the same. ★ = new combination. Unlike the stock mode, Jazz Mode voice-leads intelligently when switching chords.

Chord buttons	Plays
"C-MAJOR"	C6
"C-MINOR"	Cm9
"C-7th"	C13
"C-MAJOR" + "C-7th"	Cmaj9
"C-MINOR" + "C-7th"	Cm11
"C-MAJOR" + "C-MINOR"	Cdim7
"C-MAJOR" + "C-MINOR" + "C-7th"	C9♯5
"C-MAJOR" + "F-7th"	C9sus4
"C-MAJOR" + "F-MINOR"	C6/9
★ "C-MINOR" + "F-MAJOR"	Cm6/9
★ "C-MINOR" + "F-7th"	Cm7 ♭ 5 (half-diminished)
★ "C-7th" + "F-MAJOR"	C13 ♭ 9
★ "C-MINOR" + "G-7th"	Cm(maj9)